MIL-R-7052B
4 MAY 1959
SUPERSEDING
MIL-R-007052A(USAF)
31 MARCH 1958
USED IN LIEU OF
MIL-R-7052
30 OCTOBER 1950

MILITARY SPECIFICATION

RECEPTACLE, RELEASE, BOMB; TYPE A-2

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of bomb release receptable for operation on 28 volts direct current, designated type A-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-D-5028 — Drawings and Data Lists: Preparation Of Manufacturers' (For Production Aeronautical and Associated Equipment).

MIL-S-7742 — Screw Threads; Standard, Aeronautical.

MIL-P-7936 — Parts and Equipment, Aeronautical, Preparation For Delivery.

STANDARDS

MILITARY

MIL-STD-130 — Identification Marking of U. S. Military Property.

DRAWINGS

U. S. AIR FORCE

45B5800 — Receptacle Assembly; Bomb Release, Type A-2.

45A16556 — Base Assembly, Bomb Rack Release Connector Plug.

PUBLICATIONS

AIR FORCE-NAVY AERONAUTICAL BULLETIN

143 — Specification and Standards; Use Of.

(Copies of documents required by contractors in connection with specific procurement functions

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:51Z
Check the source to verify that this is the current version before use.

should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Preproduction.** This specification makes provisions for preproduction testing.

3.2 **Selection of specifications and standards.** Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin 143.

3.3 **Design and construction.** The bomb release receptacle shall be constructed in accordance with Air Force Drawing 45B5800 and this specification.

3.3.1 *Shock contacts.* The entering edge of the sockets shall be rounded or chamfered to guide the entrance of the pins. The socket contacts shall provide spring action for maintaining the contact pressure between pins and sockets.

3.3.1.1 *Solder cups.* The solder cups shall be such that normal soldering operation does not impair any part of the assembly.

3.3.1.2 To facilitate self-alignment of mating contacts, all socket contacts shall be designed so that the individual sockets will have an overall side-play of approximately 0.008 inch.

3.3.2 Socket contact positions shall be designated by inscribed numbers as shown on drawing 45B5800.

3.3.3 Means shall be provided for positioning of the inserts in the shell.

3.3.4 *Screw thread.* All screw threads shall be in accordance with MIL-S-7742.

3.4 **Performance.**

3.4.1 The receptacle shall mate with connector plug 45A16656, plus the function of shorting one pair of contacts (contacts 2 and 3) when the mating plug is removed from the receptacle.

3.4.1.1 The arrangement of the shorting contacts shall be normally closed. Inserting the mating plug shall separate or "open" contacts 2 and 3.

3.4.1.2 The shorting contacts shall be capable of carrying a current of 15 amperes at 28 volts dc, intermittent duty. The contacts will not be required to make or break the circuit when current is flowing

3.4.2 *Resistance of contacts.*

3.4.2.1 The resistance of the shorting contacts when closed shall be such that the potential drop shall not be greater than 70 millivolts as determined with a current flow of 15 amperes at an ambient temperature of 68° ± 5°F.

3.4.2.2 The resistance of pin and socket contacts when assembled as in service shall be such that the potential drop will not be greater than 15 millivolts as determined with a current flow of 15 amperes at an ambient temperature of 68° ± 5°F.

3.4.3 *Durability.* The bomb release receptacle assemblies shall show no electrical or mechanical defects after being subject to 500 cycles of connection and disconnection with the mating plug.

3.4.4 *Engaging and separation forces.* The force required to engage or separate individual pins from the sockets shall not exceed 3 pounds nor be less than ½ pound.

3.5 **Interchangeability** All parts having the same part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the part numbers

3.6 **Identification of product.** Equipment,

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:51Z
Check the source to verify that this is the current version before use.

assemblies and parts shall be marked for identification in accordance with MIL-STD-130. The identification data shall be as follows:

Receptacle, Bomb Release, Type A-2

Specification No. MIL-R-7052B.

3.7 Workmanship. Workmanship shall be consistent with the requirements specified herein and of the quality necessary to produce receptacles free from injurious defects

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests. The inspection and testing of receptacles shall be classified as follows:

(a) Preproduction tests.

(b) Acceptance tests.

4.3 Preproduction testing.

4.3.1 *Preproduction test sample.* A minimum of one receptacle, as specified in the contract, will be tested for approval by the procuring activity, or, when so specified in the contract, at the contractor's plant under the supervision of the procuring activity.

4.3.2 *Preproduction tests* Preproduction tests shall consist of all tests described under 4.6, Test Methods, in the same order.

4.4 **Acceptance tests.** Acceptance tests shall consist of:

(a) Individual tests.

(b) Sampling plan and tests.

4.4.1 *Individual tests.* Each receptacle shall be subjected to the following tests as described under 4.6, Test Methods, of this specification:

(a) Examination of product.
(b) Functional tests.

4.4.2 *Sampling tests.*

4.4.2.1 *Sampling plan.* Test samples shall be selected at random from receptacles that have successfully completed all individual tests. Sampling tests shall consist of all tests described under 4.6, Test Methods. Samples shall be selected in the following manner:

(a) Two receptacles from the first 100 items on the contract or order.

(b) One receptacle from each succeeding 250 or fraction thereof on the contract or order.

4.4.2.2 *Rejection and retest.* When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.4.2.3 *Individual tests may continue.* For operational reasons, individual tests may be continued pending the investigation of a sampling test failure But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of the specification.

4.4.3 *Defects in items already accepted.* The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions. Tests shall be con-

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:51Z
Check the source to verify that this is the current version before use.

ducted at normal ambient atmospheric conditions unless otherwise specified.

4.6 Test methods.

4.6.1 *Examination of product.* Each receptacle shall be examined for conformance to this specification and the applicable drawings with respect to workmanship, material, dimensions, and markings.

4.6.2 *Functional test.* The shorting contacts of each bomb release receptacle shall be tested for separation when contact pins are inserted. Before the insertion of the contact pins, current shall flow between socket contacts 2 and 3 when the contacts are placed in series with an ammeter, suitable resistance, and 28 volts direct current. Insertion of the contact pins into 2 and 3 shall interrupt the current flow.

4.6.3 *Pin and socket contact resistance.* The electrical resistance of each pin and socket contact shall be determined by measuring the potential drop across the mated pin and socket contacts while 15 amperes of direct current flow through them. The millivolt drop shall not exceed the value specified in 3.4.2 of this specification.

4.6.4 *Shorting contact resistance.* The electrical resistance of the shorting contacts shall be determined by measuring the potential drop across contacts 2 and 3 when the mating pin contacts are removed. The potential drop shall be measured at the extreme end of the solder cups while 15 amperes of direct current flow through the contacts. The millivolt drop shall not exceed the value specified in 3.4.1 of this specification.

4.6.5 *Contact engaging and separation forces.* The receptacle shall be mounted in a suitable position with provisions for applying loads. Gradually increasing loads shall be applied until the pin contact properly engages with or separates from the socket. The load required to engage or separate the individual pins from the sockets shall not exceed 8 pounds nor be less than ½ pound, as specified in 3.4.4.

4.6.6 *Durability.* The bomb release receptacle shall be subjected to 500 cycles of plug insertion and withdrawal at a rate not exceeding 600 cycles per hour. At the completion of the durability tests, the receptacle shall meet the requirements specified in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-P-7936. Preservation for level A shall be in accordance with method IA-8, MIL-P-116.

6. NOTES

6.1 **Use.** The type A-2 bomb release receptacle is intended for use on military aircraft for connecting and disconnecting the electric circuits of bomb shackle releases. In addition, the bomb release receptacle employs a feature that maintains electrical continuity through the bomb stations when the respective bomb shackle release is not installed.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation and packaging, and packing.

Notice When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:51Z
Check the source to verify that this is the current version before use.

or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Sig
Air Force

Preparing activity:
Air Force

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:51Z
Check the source to verify that this is the current version before use.